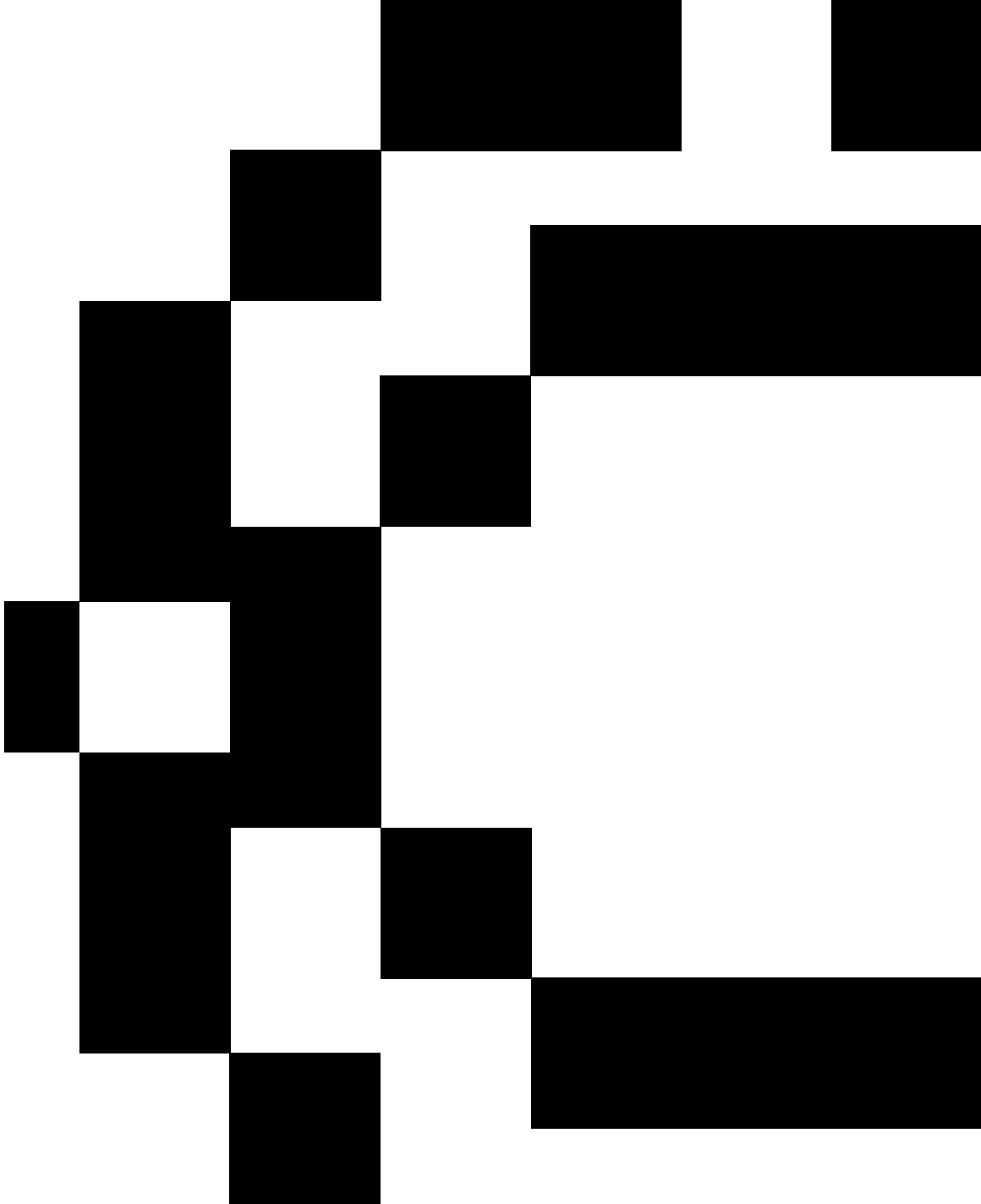
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 06 - Introduction to For Loops

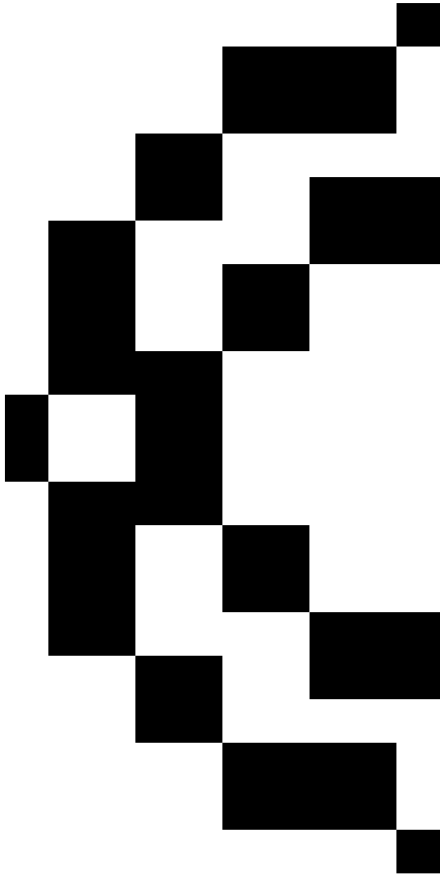


Exercise 1

Using a `for` loop, print all the integers from 1 to 10 inclusive, separated by spaces on a single line.

Example

- `Input:  (no input)`
- `Output: 1 2 3 4 5 6 7 8 9 10`

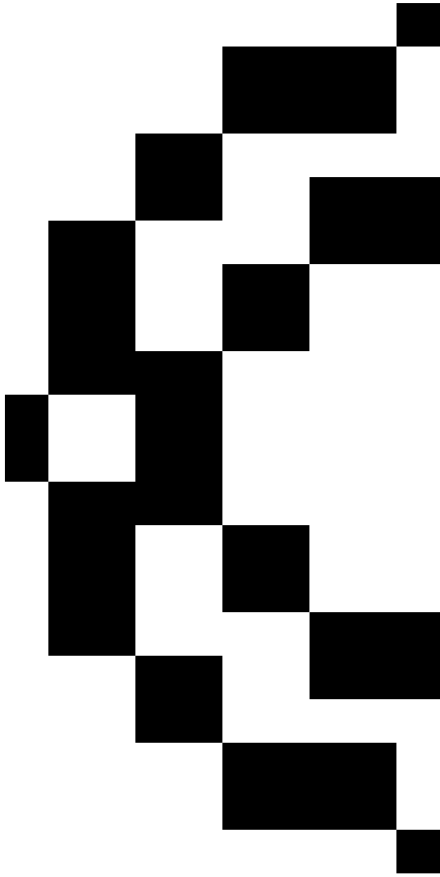


Exercise 2

Using a `for` loop, print all the numbers from 1 to 100 that are divisible by 3.

Example

```
Input:  (no input)
Output: 3
        6
        9
        ...
        99
```

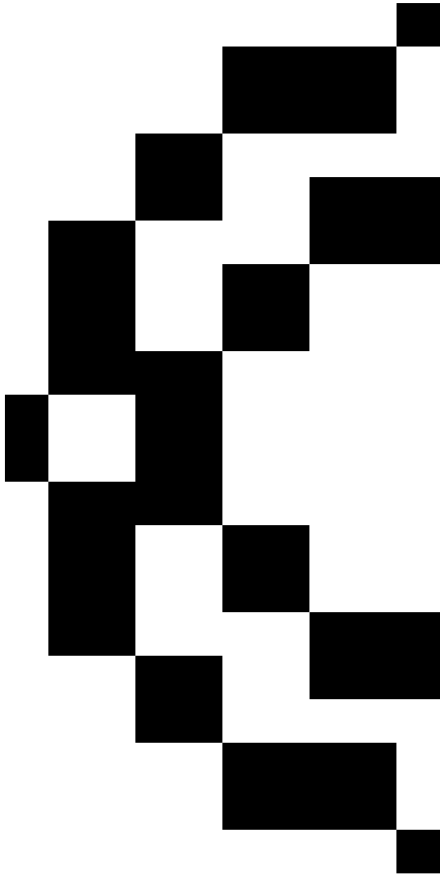


Exercise 3

Ask the user for an integer `n`, then build a list containing all the integers from 1 to `n` inclusive, using a `for` loop and `.append()`.

Example

- `Input:  n = 5`
- `Output: [1, 2, 3, 4, 5]`

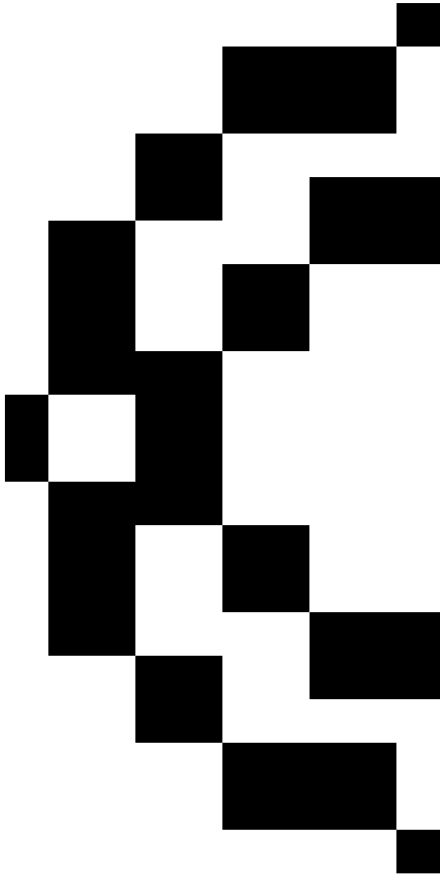


Exercise 4

Using a `for` loop, print all the numbers from 1 to 100 on a single line, separated by spaces. Whenever the running count reaches a multiple of 10, print the message `"That's 10 numbers printed!"` on its own line before continuing.

Example

```
Input:  (no input)
Output: 1 2 3 4 5 6 7 8 9 10
        That's 10 numbers printed!
        11 12 13 14 15 16 17 18 19 20
        That's 10 numbers printed!
        ...
```

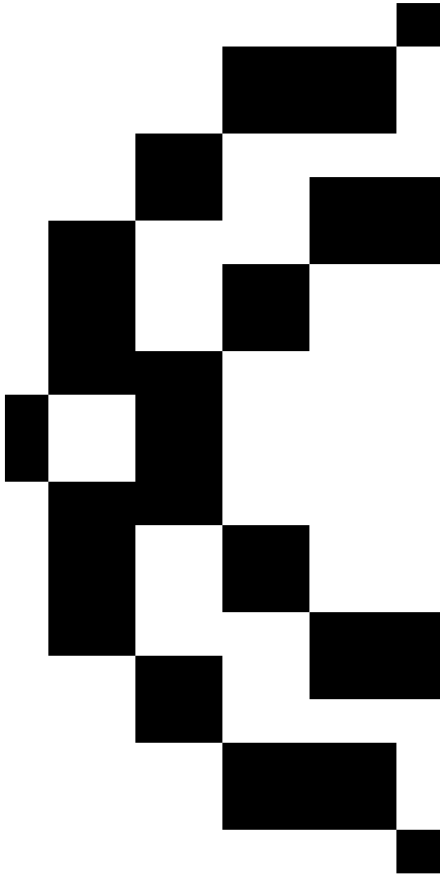


Exercise 5

Ask the user for a word. Check every letter in the word one by one, and each time the letter `"a"` is found, print `"Found one!"`.

Example

```
Input:  word = "banana"
Output: Found one!
        Found one!
        Found one!
```

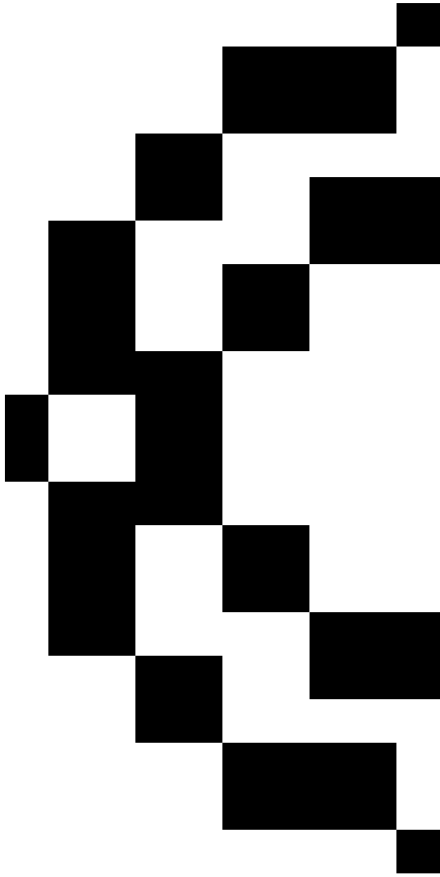


Exercise 6

Ask the user for a word and a single letter. Print how many times that letter appears in the word.

Example

- `Input:  word = "banana", letter = "a"`
- `Output: 3`

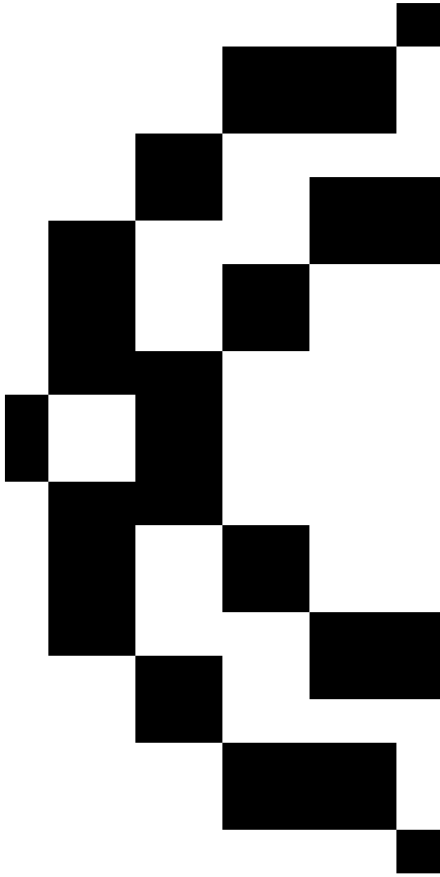


Exercise 7

Ask the user for a word. Count how many consonants (letters that are not `a`, `e`, `i`, `o`, or `u`) it contains, and print the answer.

Example

- `Input:  word = "python"`
- `Output: 4`



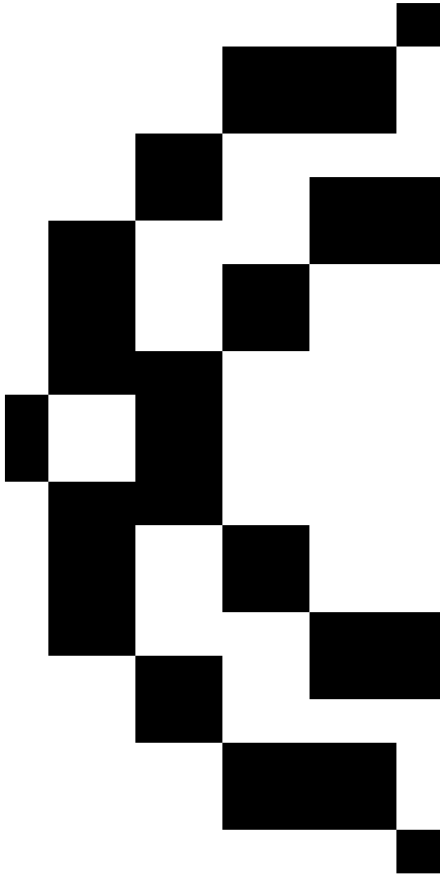
Exercise 8

Ask the user, in order: the number of classes held, the number of classes they attended, and whether they have a medical cause for absences (`"Y"` or `"N"`). A student may not sit the exam if their attendance is below 75%, unless they have a medical cause. Print their attendance percentage and whether they are allowed to sit the exam.

Examples

- `Input:  held = 20, attended = 12, medical = "Y"`
- `Output: Percentage is 60.0% with medical cause, you CAN attend the exam.`

- `Input:  held = 20, attended = 12, medical = "N"`
- `Output: Percentage is 60.0%, you CANNOT attend the exam.`



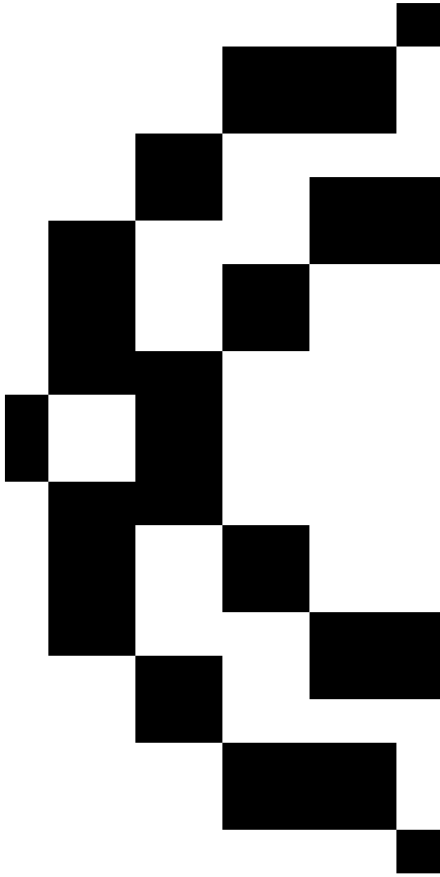
Exercise 9

Ask the user for two grades, `grade1` and `grade2`, each expected to be between 0 and 20. If either grade is outside that range, print an error message. Otherwise, ask the user to type `"Average"` or `"Greatest"`: for `"Average"`, print the average of the two grades; for `"Greatest"`, print whichever grade is larger.

Examples

- `Input:  grade1 = 14, grade2 = 17, choice = "Greatest"`
- `Output: 17.0 is the greatest`

- `Input:  grade1 = 24, grade2 = 10`
- `Output: Grade1 or Grade2 is out of the interval [0,20]`



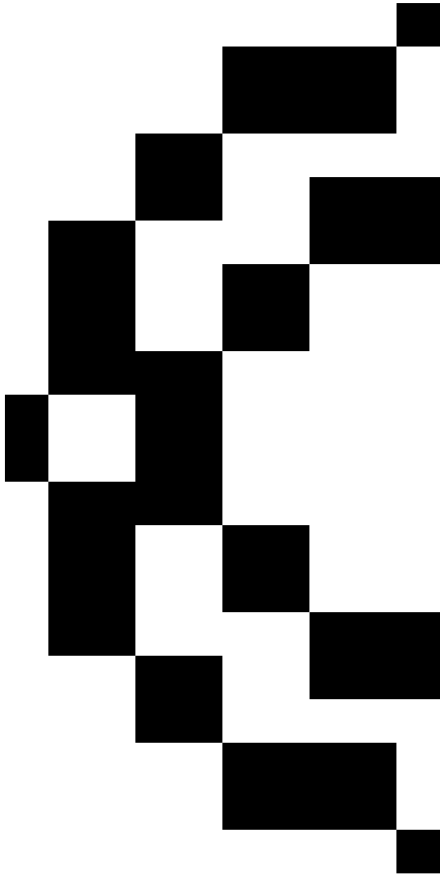
Exercise 10

A company has four departments: finance, software, management, and maintenance. Ask the user for their department and how many years they have worked there. Software and management staff get a 30% bonus with 3 or more years of experience, otherwise 25%. Finance staff get a 20% bonus with 5 or more years of experience, otherwise 15%. Maintenance staff always get a 10% bonus. Print the applicable bonus.

Examples

- `Input:  department = "software", years = 4`
- `Output: Bonus: 30%`

- `Input:  department = "finance", years = 2`
- `Output: Bonus: 15%`



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Thank you

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